

# User Agent Inspector — Field Reference

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## 1. User-Agent String

| Attribute  | Detail                                                                                                 |
|------------|--------------------------------------------------------------------------------------------------------|
| Definition | A text string sent by the browser in every HTTP request identifying the application, OS, and platform. |
| API        | <code>navigator.userAgent</code>                                                                       |

### Extraction:

```
const ua = navigator.userAgent;
```

### Example output:

Mozilla/5.0 (Macintosh; Intel Mac OS X 10\_15\_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.0.0 Safari/537.36

### Platform Variations:

| Platform            | Typical UA Pattern                                                                      |
|---------------------|-----------------------------------------------------------------------------------------|
| Desktop (macOS)     | Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/...                         |
| Desktop (Windows)   | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/...                               |
| iOS (Safari)        | Mozilla/5.0 (iPhone; CPU iPhone OS 17_5 like Mac OS X) AppleWebKit/...                  |
| iOS (Chrome)        | Same as Safari (WebKit required on iOS) + CriOS/125.0                                   |
| Android (Chrome)    | Mozilla/5.0 (Linux; Android 14; Pixel 8) AppleWebKit/... Chrome/125.0 Mobile Safari/... |
| Android (Samsung)   | Same structure + SamsungBrowser/25.0                                                    |
| iPad (Desktop mode) | Identical to macOS — use <code>maxTouchPoints</code> to differentiate                   |

**Key note:** On iOS, all browsers (Chrome, Firefox, etc.) are forced to use WebKit. Their UA strings differ only by a version token (e.g., `CriOS`, `FxiOS`), not by engine.

## 2. Browser Environment

These values come from JavaScript APIs, not the UA string.

### 2.1 Touch Support

| Attribute         | Detail                                                                          |
|-------------------|---------------------------------------------------------------------------------|
| <b>What it is</b> | How many fingers the screen can detect simultaneously. 0 means no touch screen. |
| <b>API</b>        | <code>navigator.maxTouchPoints</code>                                           |

```
const points = navigator.maxTouchPoints; // 0 = no touch, 5 = typical tablet/phone
```

| Platform                 | Typical Value |
|--------------------------|---------------|
| Desktop (no touch)       | 0             |
| iPad / iPhone            | 5             |
| Android phone/tablet     | 5 or 10       |
| Windows touch laptop     | 10            |
| Desktop + external touch | 1-10          |

**Use case:** Distinguishes iPad-in-desktop-mode ( `Macintosh` UA + `maxTouchPoints > 1` ) from a real Mac ( `maxTouchPoints === 0` ).

## 2.2 Platform

| Attribute         | Detail                                                      |
|-------------------|-------------------------------------------------------------|
| <b>What it is</b> | A short string identifying the OS/CPU the browser runs on.  |
| <b>API</b>        | <code>navigator.platform</code> (deprecated but functional) |

```
const platform = navigator.platform;
```

| Platform                    | Value                         |
|-----------------------------|-------------------------------|
| macOS (Intel/Apple Silicon) | MacIntel                      |
| iPad (desktop mode)         | MacIntel                      |
| Windows                     | Win32                         |
| Linux                       | Linux x86_64                  |
| iPhone                      | iPhone                        |
| Android                     | Linux armv81 Or Linux aarch64 |

## 2.3 Cookies

| Attribute         | Detail                                           |
|-------------------|--------------------------------------------------|
| <b>What it is</b> | Whether the browser accepts first-party cookies. |

|            |                         |
|------------|-------------------------|
| <b>API</b> | navigator.cookieEnabled |
|------------|-------------------------|

```
const enabled = navigator.cookieEnabled; // boolean
```

| Platform                    | Default                              |
|-----------------------------|--------------------------------------|
| All modern browsers         | true                                 |
| Private/incognito mode      | true (still accepts session cookies) |
| Explicitly disabled by user | false                                |

## 2.4 CPU Cores

| Attribute         | Detail                                                  |
|-------------------|---------------------------------------------------------|
| <b>What it is</b> | Number of logical CPU threads available to the browser. |
| <b>API</b>        | navigator.hardwareConcurrency                           |

```
const cores = navigator.hardwareConcurrency; // e.g., 8
```

| Platform          | Typical Values       |
|-------------------|----------------------|
| Desktop           | 4–16+                |
| iPhone / iPad     | 6 (capped by Safari) |
| Android mid-range | 4–8                  |
| Android flagship  | 8                    |

**Note:** Safari on iOS may report a lower value than the physical core count for privacy.

## 2.5 Screen Resolution

| Attribute         | Detail                                                 |
|-------------------|--------------------------------------------------------|
| <b>What it is</b> | Display size in CSS pixels + device pixel ratio (DPR). |
| <b>API</b>        | screen.width, screen.height, window.devicePixelRatio   |

```
const w = screen.width;           // CSS px
const h = screen.height;          // CSS px
const dpr = devicePixelRatio;     // physical px per CSS px
```

| Platform        | Resolution | DPR |
|-----------------|------------|-----|
| MacBook Pro 14" | 1512 × 982 | 2   |

|                     |             |       |
|---------------------|-------------|-------|
| iPhone 15 Pro       | 393 × 852   | 3     |
| iPad Pro 12.9"      | 1024 × 1366 | 2     |
| Android 1080p phone | 412 × 915   | 2.625 |
| Full HD desktop     | 1920 × 1080 | 1     |

## 2.6 Viewport

| Attribute         | Detail                                                                  |
|-------------------|-------------------------------------------------------------------------|
| <b>What it is</b> | The visible page area (excludes browser chrome, address bar, toolbars). |
| <b>API</b>        | <code>window.innerWidth</code> , <code>window.innerHeight</code>        |

```
const vw = window.innerWidth;
const vh = window.innerHeight;
```

**Note:** On mobile, viewport height changes when the virtual keyboard appears or the address bar collapses.

## 2.7 Device Memory

| Attribute         | Detail                                                            |
|-------------------|-------------------------------------------------------------------|
| <b>What it is</b> | Approximate device RAM in GB (rounded to power of 2 for privacy). |
| <b>API</b>        | <code>navigator.deviceMemory</code>                               |

```
const ram = navigator.deviceMemory; // 0.25, 0.5, 1, 2, 4, or 8
```

| Platform              | Support        |
|-----------------------|----------------|
| Chrome / Edge / Opera | Yes            |
| Firefox               | No (undefined) |
| Safari (iOS/macOS)    | No (undefined) |
| Samsung Browser       | Yes            |

## 2.8 Color Depth

| Attribute         | Detail                                     |
|-------------------|--------------------------------------------|
| <b>What it is</b> | Bits per pixel the display uses for color. |
| <b>API</b>        | <code>screen.colorDepth</code>             |

```
const bits = screen.colorDepth; // 24 or 30
```

| Value | Meaning                                           |
|-------|---------------------------------------------------|
| 24    | Standard True Color (8 bits × RGB) — 16.7M colors |
| 30    | HDR / Deep Color (10 bits × RGB) — 1B colors      |

Consistent across all platforms; value depends on hardware, not OS.

## 2.9 Reduced Motion Preference

| Attribute  | Detail                                                                                |
|------------|---------------------------------------------------------------------------------------|
| What it is | User has enabled "Reduce Motion" in OS accessibility settings to minimize animations. |
| API        | <code>window.matchMedia("(prefers-reduced-motion: reduce)")</code>                    |

```
const reduced = matchMedia("(prefers-reduced-motion: reduce)").matches; // boolean
```

| Platform | Where to enable                                                   |
|----------|-------------------------------------------------------------------|
| iOS      | Settings → Accessibility → Motion → Reduce Motion                 |
| Android  | Settings → Accessibility → Remove animations                      |
| macOS    | System Settings → Accessibility → Display → Reduce motion         |
| Windows  | Settings → Accessibility → Visual effects → Animation effects OFF |

## 2.10 Network Information

| Attribute  | Detail                                                        |
|------------|---------------------------------------------------------------|
| What it is | Estimated connection quality (not actual WiFi/cellular type). |
| API        | <code>navigator.connection</code> (Network Information API)   |

```
const conn = navigator.connection;
if (conn) {
  conn.effectiveType; // "slow-2g" | "2g" | "3g" | "4g"
  conn.downlink;      // Mbps (estimated)
  conn.rtt;            // ms (round-trip)
}
```

| Platform                        | Support |
|---------------------------------|---------|
| Chrome / Edge / Opera / Samsung | Yes     |
| Firefox                         | No      |
| Safari (iOS/macOS)              | No      |

| Effective Type | Meaning                            |
|----------------|------------------------------------|
| 4g             | $\geq 700$ Kbps, $\leq 270$ ms RTT |
| 3g             | $< 700$ Kbps, $> 270$ ms RTT       |
| 2g             | $< 70$ Kbps, $> 1400$ ms RTT       |
| slow-2g        | $< 50$ Kbps, $> 2000$ ms RTT       |

### 2.11 JavaScript Status

| Attribute  | Detail                                                       |
|------------|--------------------------------------------------------------|
| What it is | Confirms JS is running. If this code executes, it's enabled. |
| API        | Implicit                                                     |

```
// Always true if code reaches this point
const jsEnabled = true;
```

For fallback when JS is off, use `<noscript>` in HTML.

### 3. Summary

| #  | Field             | API                                    | iOS           | Android       | Desktop            |
|----|-------------------|----------------------------------------|---------------|---------------|--------------------|
| 1  | User-Agent        | navigator.userAgent                    | ✓             | ✓             | ✓                  |
| 2  | Touch Support     | navigator.maxTouchPoints               | ✓ (5)         | ✓ (5–10)      | ✓ (0 or 1–10)      |
| 3  | Platform          | navigator.platform                     | ✓             | ✓             | ✓                  |
| 4  | Cookies           | navigator.cookieEnabled                | ✓             | ✓             | ✓                  |
| 5  | CPU Cores         | navigator.hardwareConcurrency          | ✓<br>(capped) | ✓             | ✓                  |
| 6  | Screen Resolution | screen.width/height + devicePixelRatio | ✓             | ✓             | ✓                  |
| 7  | Viewport          | window.innerWidth/innerHeight          | ✓             | ✓             | ✓                  |
| 8  | Device Memory     | navigator.deviceMemory                 | ✗             | ✓<br>(Chrome) | ✓<br>(Chrome/Edge) |
| 9  | Color Depth       | screen.colorDepth                      | ✓             | ✓             | ✓                  |
| 10 | Reduced Motion    | matchMedia("prefers-reduced-motion")   | ✓             | ✓             | ✓                  |

|    |              |                                   |   |               |                    |
|----|--------------|-----------------------------------|---|---------------|--------------------|
| 11 | Network Info | <code>navigator.connection</code> | ✗ | ✓<br>(Chrome) | ✓<br>(Chrome/Edge) |
| 12 | JavaScript   | Implicit                          | ✓ | ✓             | ✓                  |

## 4. References

1. MDN — Navigator API: <https://developer.mozilla.org/en-US/docs/Web/API/Navigator>
2. MDN — User-Agent: <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/User-Agent>
3. W3C — Network Information API: <https://wicg.github.io/netinfo/>
4. W3C — Device Memory: <https://www.w3.org/TR/device-memory/>